

28 Oct '25
BPwise
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PRE-MIP has 17 nodes & 472 serials

$$T_A = \sum_{i=1}^{472} a_i = 173,443.3 \text{ ft}^2$$

$$\mu(a) = 367.4646 \text{ ft}^2$$

will use 475 serials (5 x 95 each)

$$475 * \mu_a = 174,545.7$$

If we have scale-free $1/f$ distribution

$$P(a \leq f \leq b) = \ln(b) - \ln(a) = P(2a \leq f \leq 2b)$$

size-shrink TYPE ft^2

$c(\ln 2 - \ln 1) = .69$	A	710.8
$c(\ln 3 - \ln 2) = .41$	A	415.8
$c(\ln 4 - \ln 3) = .29$	C	295.0
$c(\ln 5 - \ln 4) = .22$	d	228.8
$c(\ln 6 - \ln 5) = .18$	E	187.0

INDIVIDUAL serials

MEAN of TYPE $\pm 10\%$

so TYPE-C are
dist $U[265, 325]$
i.e. 295 ± 30

$$\sum (95a) =$$

$$174,553 \text{ ft}^2$$

1500 miles $\approx$ HAVANA $\leftrightarrow$ TRINIDAD & TOBAGO

Need ≈ 18 LOCATIONS e.g. 4 origins

6 intermediate

8 final

the 8 final can stay same as
Data 2

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all the following
data is very approximate.

THUS, reasonable variations, such
as $\pm 25\%$ for # of vehicles and $\pm 10\%$
for cargo ~~to~~ area and various size
~~classes~~ types, should be used in
algorithm design & test.

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ZERO C-17 or C-5

in PRE-MIP's "2 MLR" scenario

(1 MLR assumed to already be in place)

For each transport
we need 6 items

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SPEED

1 way

RANGE COMBAT RANGE $\geq$ 1 way / 2

deck Area

available

domain (LAND, SEA)

AVAILABILITY% (e.g. 10/24 ~~h~~ hours/day)

RANGE $\swarrow$ COMBAT RADIUS ≈ 1631 miles

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BPCWise

KC-130: ~ 3262 miles @ 400 MPH

Cargo deck: $41 \times 10 \text{ ft} = 410 \text{ ft}^2 < 710$

~~RANGE~~

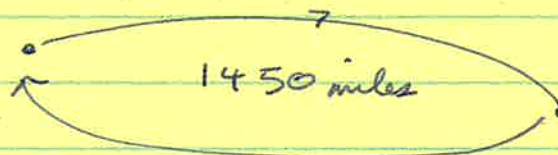
$$\frac{174,553}{410} = 426 \text{ trips } (425.74)$$

are chore
only INDIVIDUAL
SERIALS which
cannot fit in
KC-130?

WITM ~ 40 KC-130 $\rightarrow$ 10.65 trips each
in PRE-MIP

I.E. all 40 make 10 trips,
26 make on 11th

Timing:



$$\frac{2900 \text{ miles}}{400 \text{ MPH}} * 11 = 79.75 \approx 80 \text{ hours} = 3.3 \text{ days}$$

$\nwarrow$ 7.25 hr flight time

DOUBLE that for MAINT, LOAD, UNLOAD, @ ~~2.14~~

Peter B: 10 hrs/day usable

$$\frac{24}{10} \times 80 \text{ hr} = 192 \text{ hrs} = 8 \text{ days}$$

$$(3.323 \frac{24}{10} = 7.975 \checkmark)$$

KC-130 requires 2500 ft runway (^{MAYBE} 3127, 953m loaded)

GUADALOUPE: 3125 m, BVI: 1415 m, TOT: 9002 ft

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LAW: ~ 22 frats ≈ 25.3 mphrange 6500 nm ≈ 7480 statute statute miles
350-400 ft long (1/2 is usable deck)GUESSING $w = L/5$ from pictures27 LAWS (2 /MLR) $L \approx 375$, $w \approx 75$

$$\frac{375}{2} \times 75 \approx 14,062.5 \text{ ft}^2/\text{LAW}$$

$$\text{as } u(\text{area}) \approx 367.5 \quad \frac{174,553 \text{ ft}^2}{14,062.5 \text{ ft}^2/\text{LAW}} = 12.4 \text{ LAW}$$

each LAW can hold average of 38.3 serials

size type

A	19
B	33
C	47
D	61
E	75

or 33 of B and 1 of C or D or E etc.

$$\frac{1450 \text{ miles}}{25.3 \text{ mph}} = 57.3 \text{ hours one way}$$

So "1 Big Boat" or a dozen in 11
~~is far~~ might be faster than relays
 of KC-130

7/d

MV-22 (MOSTLY PAX: Marines + equip)

24 MV-22 (2 SQDN of 12 each)

Range = 3892 km 1 way

1182 km combat radius ≈ 732.8 mi

@ 448-473 KPH

≈ 460.9 KPH

≈ 285.5 MPH

Cargo L = 24.17"

bay W = 5.92"

A = 143.0864 ft²

≈ 143 ft²

< 187 ft² TYPE-E

MV-22 designed for 12 USMC passengers + SUPPLIES, etc
or about 11.9 ft²/PAX

1 MLR $\approx$ 1,800 ~ 2000 Marines
1400

2 MLR $\approx$ 3800 Marines

$\frac{3800}{12} = 300 \sim 394$ MV22 flights
 ≈ 317

$\frac{317}{24} = 11.7$ flights each

(but ONLY 733 miles,
1/2 way from H to TBT)

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"LIMITED" # " & CH-53K

"LIMITED" compared to L4 = ? 8-10 SWAG

CRUISE $\approx$ 196 mph

36,000 MAX LIFT (much less for decent range)

32 PAX

CABIN L 30' 0"
W 8' 7" 8.5833'
A = 257.5 ft²

COMBAT RADIOS $\approx$ 460 miles

1 way Range $\approx$ 920 miles

4 ORIGINS:

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28 Oct 25
BP cruise

CANCUN, MIAMI, HAVANA, SANTA CLARA
CUBA

8 DESTINATIONS: FIRST SIX
as per DATA 2:
USVI, DOMINICA, A&B, MARTINIQUE, LUCIA, ST. VINCENT

SHORTEST range = 460 C. Rad. of CH-53

HAVANA → JAMAICA, ← KINGSTON ≈ 450 mi ← NEW
J → D.R. ≈ 450 mi
DR → ARUBA ≈ 450 ← NEW

USVI

ANTIGUA & BARBUDA

2 S. DOMINGO (D. Republic)

4 S. JUAN (Puerto Rico)

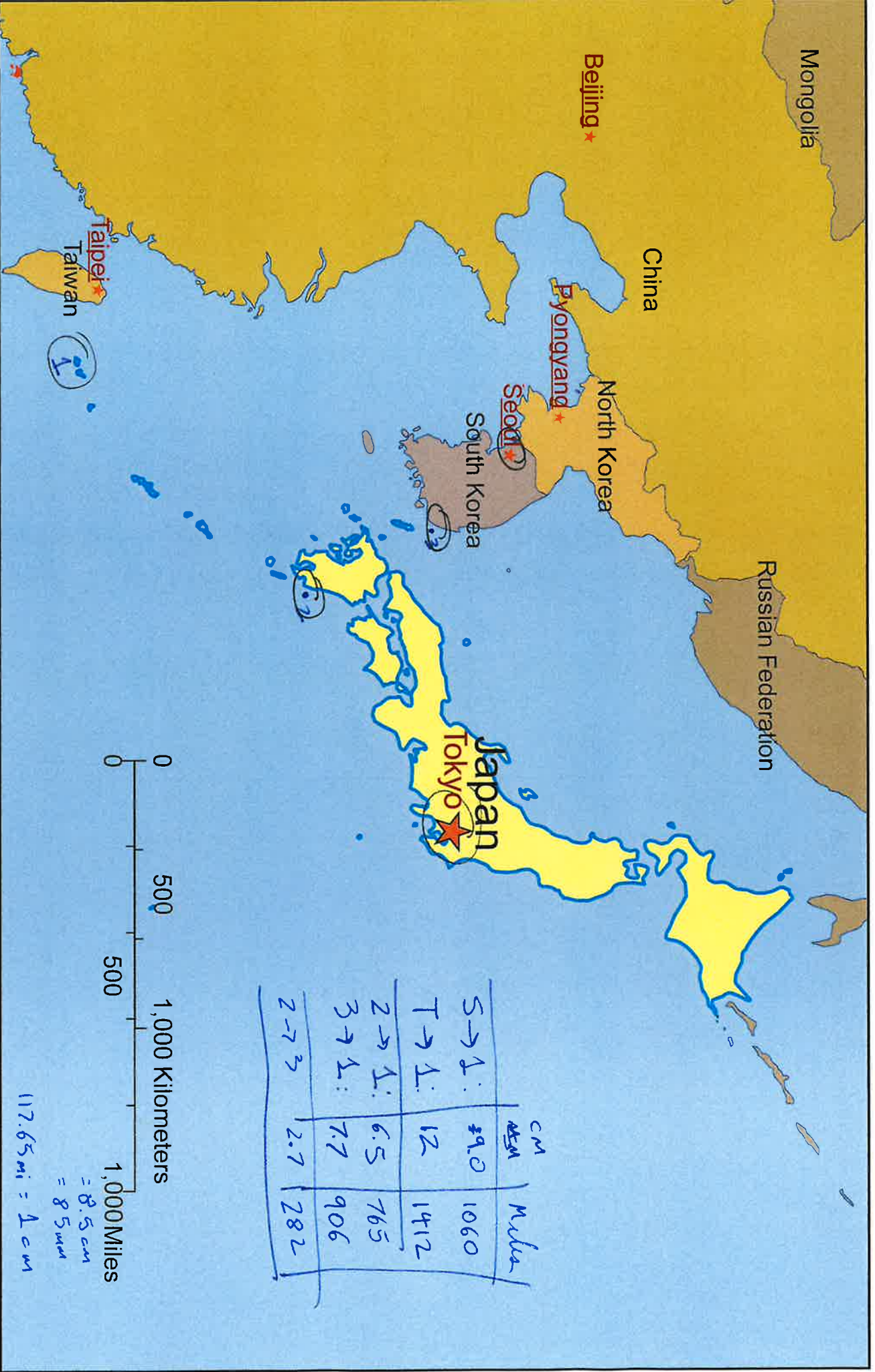
5 GUADELOUPE (FR)

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Need 1 more Intermediate and 2 more final
Margarita

159 km ON Trinidad @ 100460

413 km ON BARBADOS



TROPIC OF CANCER



1412 mi = 18.8 cm $\approx$ HAWAII - GUADALUPE
T RIMMAD - SATRA
Q TOEBO - CHRA